

- 9 The photograph shows a moose.

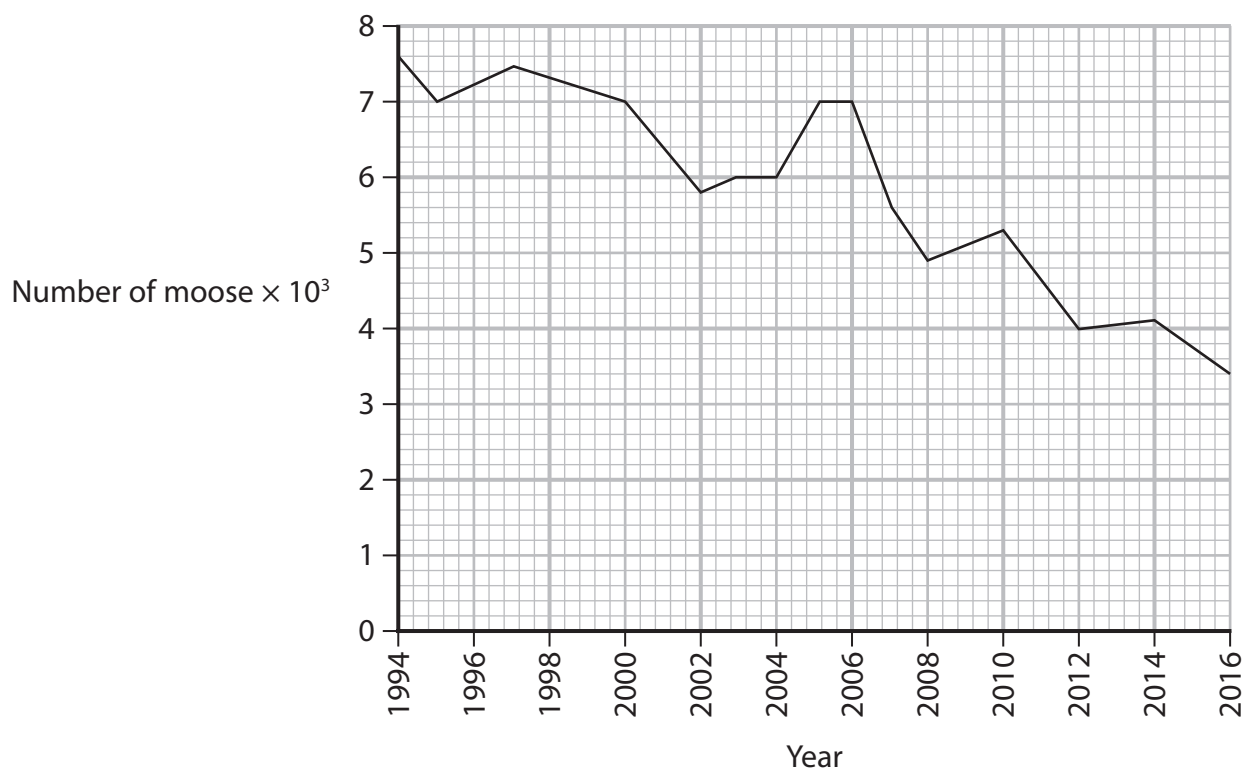


(Source: © blickwinkel/Alamy Stock Photo)

Moose are large mammals that eat grass and the shoots of plants.

The number of moose is declining in parts of America.

- (a) The graph shows the number of moose in one part of America from 1994 to 2016.



- (i) In 2016, the moose were not counted. The value shown on the graph is an estimate.

Describe how these data were used to estimate the number of moose in 2016.

(1)

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(ii) Calculate the percentage decrease in the number of moose from 1994 to 2016.

(2)

Answer %

(iii) It has been suggested that global warming is responsible for the decrease in the number of moose from 1994 to 2016.

Sketch a graph to show this relationship.

(3)



(b) Ticks are parasites that live in the fur of moose.

Ticks are thought to be responsible for the decline in the number of moose.

The table shows the frequency distribution of ticks in a population of moose.

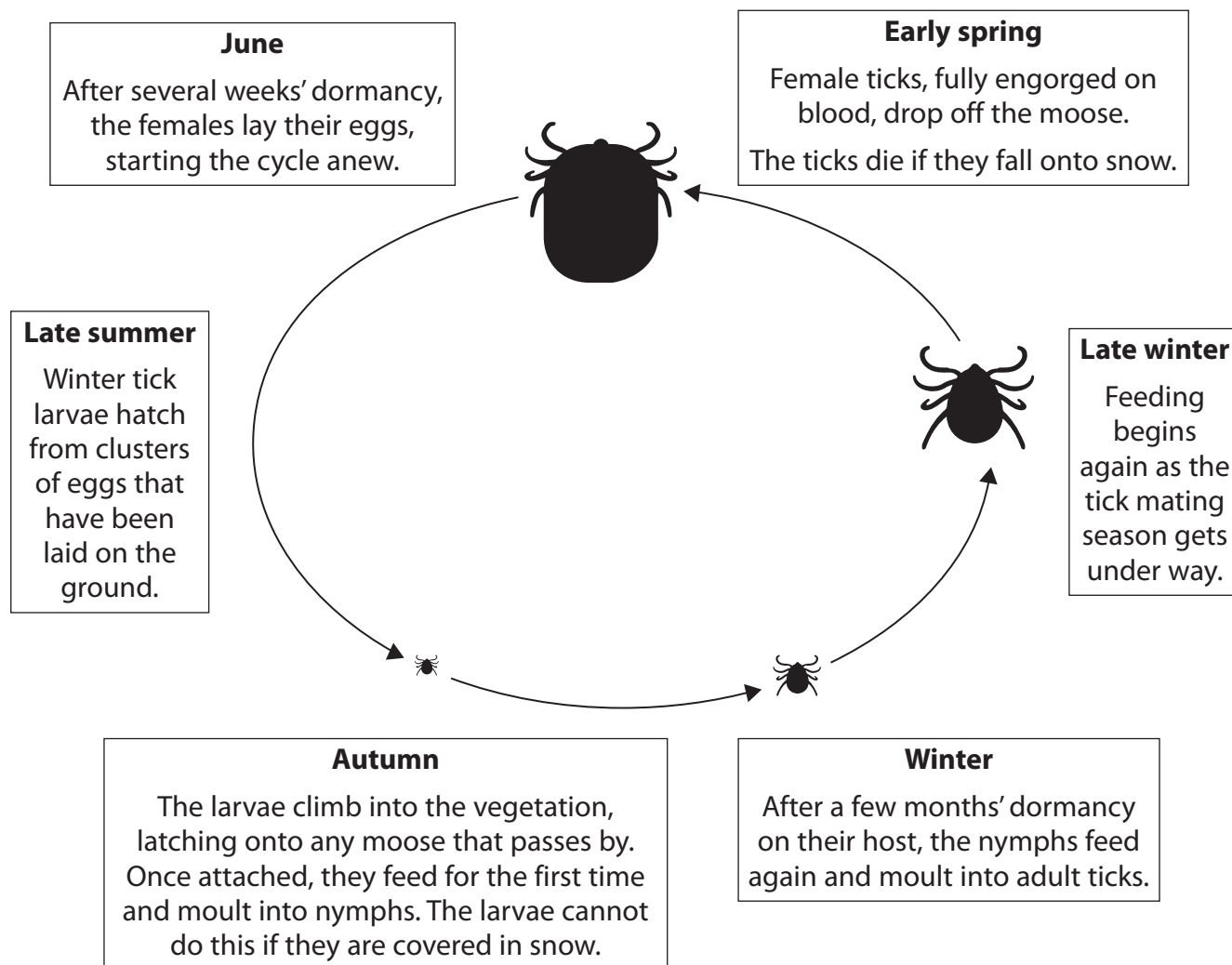
Number of ticks	Number of moose
1 to 9 999	32
10 000 to 19 999	46
20 000 to 29 999	41
30 000 to 39 999	32
40 000 to 49 999	22
50 000 to 59 999	15
60 000 to 69 000	9
70 000 to 79 999	4
80 000 to 89 999	4
90 000 to 99 999	3
100 000 or more	6

- (i) Calculate the percentage of moose that have 50 000 or more ticks on their bodies.

(2)

Answer %

*(ii) The diagram shows the life cycle of the tick.



The ticks feed on the blood of the moose. One tick can remove 200 to 600 times its body mass.

The ticks irritate the moose. The moose will scratch against trees causing large clumps of fur to fall off.

Explain how global warming might affect the life cycle of ticks and result in the decline in the number of moose.

Use all the information in the question to support your answer.

(6)

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(Total for Question 9 = 14 marks)

TOTAL FOR PAPER = 90 MARKS